



Anxiolytic effect of a herbal medicine, *yokukansan*, in aged rats: Involvement of serotonergic and dopaminergic transmissions in the prefrontal cortex

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ABSTRACT

Aim of the study: Aging is thought to affect emotions including anxiety, and a herbal medicine, *yokukansan* (YKS), is used to treat emotional disturbances associated with age-related neurodegenerative disorders such as Alzheimer's disease, but its pharmacological properties have not been fully understood. The present study was designed to examine whether YYS improves age-related anxiety using F344/N aged rats. Moreover, the effects of YYS on liver function were examined.

Materials and methods: YYS was administered to 21-month-old aged rats for 3 months. Locomotor activity of young control (4 months old), aged control (24 months old), and YYS-treated aged rats was examined, and the anxiety-related responses of these animals were evaluated by counting the number of excrements during locomotor activity measurement and in the elevated plus-maze test. The extracellular concentrations of serotonin and dopamine in the prefrontal cortex (PFC) were also measured using a microdialysis technique. Moreover, concentrations of aspartate aminotransferase (AST), alanine aminotransferase (ALT), and ammonia (NH₃) in plasma were measured.

Results: Although locomotor activity did not change among any experimental groups, the number of excrements was significantly increased in aged rats compared to young rats, and this increase was significantly improved by YYS treatment. Aged rats also showed significant decreases in time and frequency in the open arm of the elevated plus-maze, and these decreases were significantly improved by YYS treatment. Extracellular concentrations of serotonin and dopamine in the aged PFC were significantly decreased; serotonin was increased over the level of young rats and dopamine was partially improved by YYS treatment, respectively. In addition, YYS improved age-related increase in NH₃ concentration, but did not affect AST and ALT.

Conclusions: YYS has improving activity for age-related increased anxiety and enhances serotonergic and dopaminergic transmissions in the aged PFC. These mechanisms provide information important for the treatment of anxiety in the elderly. Furthermore, the present data confirm partially the *Kampo* concept "liver disease".

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1. Introduction

Approximately 10% of the world population is suffering from one or many forms of anxiety disorders, for example, panic attacks, social phobias, or generalized anxiety disorders (WHO, 2001). In addition, anxiety is sometimes a problem in several age-related neurodegenerative disorders, such as Alzheimer's disease, and even in the healthy elderly. The main therapeutic approach involves the use of benzodiazepine receptor agonists and selective serotonin reuptake inhibitors (SSRIs) (Hoehn-Saric, 1982; Hardman et al., 2001). On the other hand, drugs obtained from natural plant

sources are considered to be safe alternatives for the treatment of neuropsychiatric disorders (Kienzle-Horn, 2002; Carlini, 2003), and the use of herbal medications has tended to increase (Beaubrun and Gray, 2000). Indeed, several traditional herbal medicines (called *Kampo* drugs in Japan, remedies composed of specified mixtures of crude drugs derived from plant, animal, and mineral materials) are effective in the field of neuropsychiatry (Kanba et al., 1998). A *Kampo* drug *yokukansan* (YKS, *yi-gan san* in Chinese) has been widely used in a variety of clinical situations for treating several symptoms associated with age-related neurodegenerative disorders, such as behavioral and psychological symptoms of dementia (BPSD) (Iwasaki et al., 2005; Mizukami et al., 2009; Monji et al., 2009) and sleep disturbance in patients with dementia (Shinno et al., 2007), with very few side effects. Although scientific evidence of the clinical effects of this drug is limited, several basic researches

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have demonstrated that YKS improves learning disturbances and aggression in an animal model of Alzheimer's disease (Sekiguchi et al., 2009; Tabuchi et al., 2009) and ameliorates beta-amyloid-induced neurotoxicity *in vitro* (Tateno et al., 2008). These findings have suggested that YKS is effective for emotional disturbances in age-related neurodegenerative disorders.

Thus, YKS has psychotropic action that is involved in the improvement of several psychiatric symptoms. Considering that aging is thought to affect emotion, including anxiety, it is possible that YKS can modulate emotion in aged subjects. Nevertheless, there is no scientific evidence in the literature about its potential pharmacological effects on age-related emotional disturbances. The present study was undertaken to investigate whether YKS can improve age-related emotional changes. For this purpose, we compared the anxiety state of aged rats with young rats by assessing changes in the number of excrements when rats were placed in a novel environment and in the elevated plus-maze test. In addition, the influence of aging and the effects of YKS treatment on serotonergic and dopaminergic transmissions in the prefrontal cortex (PFC) were examined to explore the mechanisms underlying age-related anxiety and anxiolytic activity of YKS. Originally, YKS has been used for many years to treat "liver disease" that is a concept in traditional Japanese (*Kampo*) or Chinese medicine and that means not only emotional disturbances, including irritability, susceptibility to rage, excitability, and anxiety, but also dysfunction of the liver itself. Therefore, we further measured concentrations of aspartate aminotransferase (AST), alanine aminotransferase (ALT), and ammonia (NH₃) in plasma to diagnose liver function and to confirm the "liver disease" concept.

2. Materials and methods

2.1. Animals

All animal experiments were performed in accordance with our institutional guidelines after obtaining permission from the Laboratory Animal Committee at the National Center for Geriatrics and Gerontology. The experiments in the present study were designed to minimize the number of animals used and to minimize their suffering.

Naive male F344/N rats were used. Four-month-old rats weighing 270–300 g were purchased from Japan SLC (Shizuoka, Japan) and used as young rats. Twenty-one-month-old rats weighing 400–450 g were obtained from the Aging Farm, which was established at our institute in 2000 (Tanaka et al., 2000), and used as aged rats. Aged rats were produced by keeping for 18 months in the Aging Farm after purchasing from the same breeder at 3 months old. They were housed two per cage in a temperature (23 ± 1 °C) and light-controlled (12 h light/dark schedule; lights on at 8:00 a.m.) environment and were fed laboratory food and water *ad libitum*.

2.2. Drug administration

YKS was supplied in the form of a water-extracted dried powder manufactured from a mixture of the crude drugs (Tsumura Co., Tokyo, Japan) (see Table 1). The concentration of several effective chemicals was defined for each crude drug as an internal standard in the company's guide to Good Manufacturing Practices. The powdered drug was mixed at a concentration of 3% (w/w) with food pellets, made by Japan Clea (Tokyo, Japan).

Aged rats were divided into two groups. One group was fed standard pellet chow for rodents (CE-2; Japan Clea) as the control group, and the other group was fed CE-2 containing YKS as the treatment group. The feeding period was 3 months; thus, at the end of the feeding, aged rats were 24 months old.

Table 1

Crude drug composition of yokukansan (YKS).

Plant name	Composition (g)	Major components
<i>Atractylodis Lanceae Rhizoma</i>	4.0	Atractylodin
<i>Poria</i>	4.0	Eburicoic acid
<i>Cnidii Rhizoma</i>	3.0	Cnidilide
<i>Uncariae Uncis cum Ramulus</i>	3.0	Rhynchophylline
<i>Angelicae Radix</i>	3.0	Ligustilide
<i>Bupleuri Radix</i>	2.0	Saikosaponin a, c, d, e
<i>Glycyrrhizae Radix</i>	1.5	Glycyrrhizin

2.3. Locomotor activity

On the day of the experiment, the spontaneous locomotor activity of each rat under a novel environment was examined. Briefly, a rat was transferred from the home cage to the center of another plastic cage (41 cm length × 26 cm width × 21 cm height of wall) and activity was measured for 5 min using a passive infrared sensor (Supermex, PYS-001; Muromachi Kikai Co., Tokyo, Japan). After the measurement, the number of excrements was counted. The intensity of illumination of the behavior-examination room was 740 lux.

2.4. Elevated plus-maze test

This test has widely been validated to measure anxiety in rodents (Pellow et al., 1985; Lister, 1987; Mizoguchi et al., 2009a). Briefly, the elevated plus-maze was a four-arm maze made of 3-mm-thick gray vinyl chloride and clear plastic. It consisted of two open arms (50 cm length × 15 cm width, each), two enclosed arms with gray walls (50 cm length × 10 cm width × 40 cm height of wall, each) and a central platform (13 cm × 13 cm), arranged in such a way that the two arms of each type were opposite to each other. The maze was elevated 60 cm above the floor. The day after locomotor activity measurement, each rat was placed in the center of the maze, facing one of the enclosed arms. During the 5 min test period, the time spent and frequency of entering the open arms were recorded. Entry into an open arm was defined when the animal placed their two front paws into the arm.

2.5. Microdialysis

Extracellular concentrations of serotonin and dopamine in the PFC were measured using an *in vivo* microdialysis technique in freely moving animals according to our previous report (Mizoguchi et al., 2009b). Briefly, rats were stereotactically implanted with a guide cannula (CX-G; long, 6 mm; outer diameter, 0.5 mm; Eicom, Kyoto, Japan) with a stylet in the prelimbic cortical region of the PFC (anteroposterior, +3.2; lateral, +0.8; vertical, −3.0 from the bregma) (Paxinos and Watson, 2007) under pentobarbital anesthesia (50 mg/kg for young rats; 30 mg/kg for aged rats, i.p.). The rats were initially treated with xylocaine (AstraZeneca PLC, London, UK) to minimize pain, and were monitored on a daily basis for signs of distress or infection. Seven days after surgery, the rats were restrained gently while the stylet was removed and replaced with a microdialysis probe (CX-I; tip length, 2 mm; tip diameter, 0.22 mm; Eicom). Using a microinfusion pump, Ringer's solution (in mM: Na, 147; K, 4.0; CaCl₂, 3.0) was perfused at a rate of 1 µl/min. After an equilibration period of 3 h, the perfusate was collected every 30 min. To examine the response to stimuli, KCl concentration in Ringer's solution was increased to 100 mM. The concentrations of serotonin and dopamine in each perfusate (30 µl) were measured subsequently using an HPLC system equipped with a coulometric electrochemical detector (HTEC-500; Eicom). A reverse-phase ODS column (PP-ODS, 4.6 mm × 30 mm; Eicom) was used to separate each neurotransmitter with a mobile phase consisting of 100 mM sodium phosphate (pH 6.0), 900 mg/l sodium 1-decanesulfonate, and 50 mg/l EDTA.

At the end of the experiments, we histologically verified the placement of the cannula in the PFC; the cannula was placed correctly within the cingulate or prelimbic regions of all animals (Fig. 3).

2.6. Measurement of AST, ALT, and NH_3 in plasma

The day after the behavioral test, animals were sacrificed by decapitation, and the trunk blood was collected in a 15 ml polypropylene tube containing 250 μl of 100 mM EDTA-2Na (pH 7.4). The plasma was separated by centrifugation at $1800 \times g$ for 20 min, and stored at -20°C until measurement. The concentrations of AST and ALT in plasma were measured by the consensus method of Japan Society of Clinical Chemistry (JSCC) (Kotani et al., 1994), and of NH_3 were measured by the standard enzymatic method using nicotinamide adenine dinucleotide synthetase (NADS) and glucose-6-phosphate dehydrogenase (G-6-PDH). The data are expressed as IU/l for AST and ALT; $\mu\text{g/dl}$ for NH_3 .

2.7. Statistics

All data were initially analyzed using one-way analysis of variance (ANOVA). Individual between-group comparisons were employed using Fisher's protected least significant difference test.

3. Results

3.1. Locomotor activity and excrements

The influence of aging and the effects of YKS treatment on locomotor activity and the number of excrements were examined, and the results are shown in Fig. 1. Locomotor activity was not different between young and aged rats, and YKS treatment did not affect activity; however, the number of excrements counted during locomotor activity measurement was significantly increased in aged rats compared to young rats [$F(2,30) = 18.104$, $p < 0.001$], and this increase was significantly improved by YKS treatment [$F(2,30) = 18.104$, $p < 0.01$].

3.2. Elevated plus-maze test

The influence of aging and the effects of YKS treatment on the time spent and frequency of entering the open arms of the elevated plus-maze apparatus were examined, and the results are shown in Fig. 2. The time spent in the open arms was significantly less in

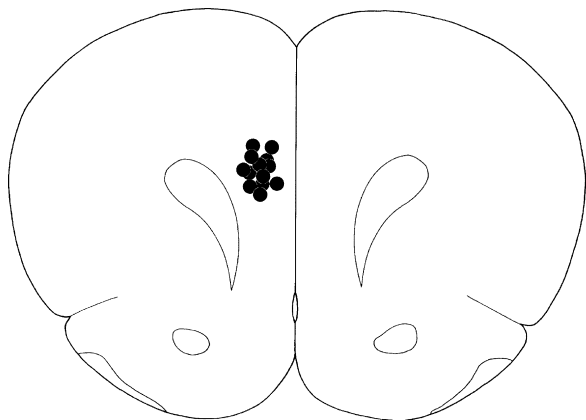


Fig. 1. Influence of aging and effect of YKS treatment on locomotor activity (A) and the number of excrements during locomotor activity measurement (B). Each column is the mean $\pm$ SEM of 10–12 rats per group. Asterisks indicate a significant difference from young control rats, $***p < 0.001$. Daggers indicate a significant difference from aged control rats, $^{\dagger}p < 0.01$.

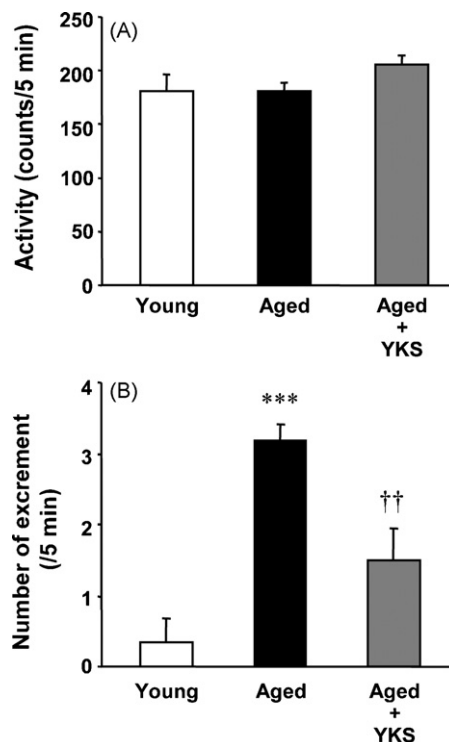


Fig. 2. Influence of aging and effect of YKS treatment on the time spent (A) and frequency (B) entered in the open arms of the elevated plus-maze. Each column is the mean $\pm$ SEM of 10–12 rats per group. Asterisks indicate a significant difference from young control rats, $***p < 0.001$. Daggers indicate a significant difference from aged control rats, $^{\dagger}p < 0.05$ and $^{\dagger\dagger}p < 0.001$.

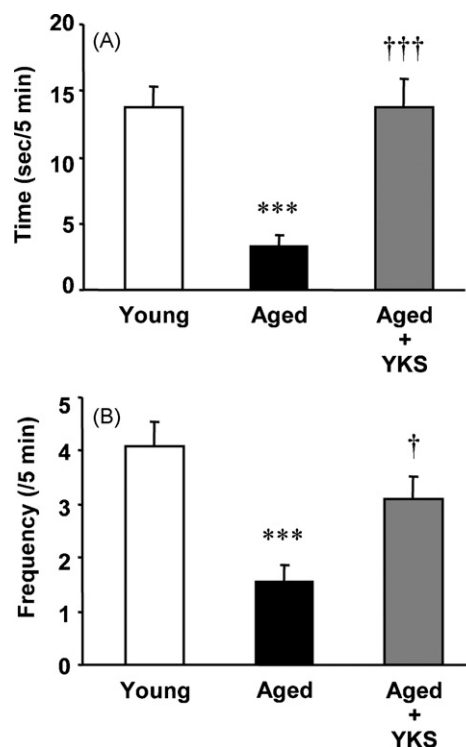


Fig. 3. Location of guide cannula tips in the prelimbic cortical region of the PFC. The microdialysis probe reached -5.0 mm (dorsoventral).

aged rats compared to young rats [$F(2,30) = 15.235$, $p < 0.001$]. Similarly, the frequency was also decreased in aged rats [$F(2,30) = 9.904$, $p < 0.001$]. In the drug treatment study, YKS significantly improved both the shorter time and decreased frequency [$F(2,30) = 15.235$, $p < 0.001$; $F(2,30) = 9.904$, $p < 0.05$, respectively].

3.3. Extracellular concentrations of serotonin and dopamine

Extracellular concentrations of serotonin and dopamine in the PFC under basal and KCl-stimulated conditions were evaluated using an *in vivo* microdialysis technique, and the concentrations were compared among young, aged, and YKS-treated aged rats (Fig. 4).

The basal serotonin concentration (Fig. 4A) was significantly decreased in aged rats compared to young rats at several time points [−90 min, $F(3,26) = 5.991$, $p < 0.05$; −60 min, $F(3,26) = 5.664$, $p < 0.05$; 0 min, $F(3,26) = 4.182$, $p < 0.05$]. The average serotonin concentrations (in pg/30 μ l, mean $\pm$ SEM) calculated from the values from −90 to 0 min in young and aged rats were as follows: 8.54 ± 0.54 and 3.93 ± 0.18 , respectively. A marked increase in the serotonin concentration was observed in young rats under perfusion of high KCl (100 mM); however, this high KCl-evoked increase was significantly attenuated in aged rats compared to young rats [30 min, $F(3,26) = 8.544$, $p < 0.001$; 60 min, $F(3,26) = 7.979$, $p < 0.01$; 90 min, $F(3,26) = 4.551$, $p < 0.05$]. Thus, the serotonin concentration in the PFC was decreased in aged rats. In the drug treatment study, both the decreased basal concentration and attenuated KCl-evoked response in the aged PFC were significantly improved by YKS treatment [−90 min, $F(3,26) = 5.991$, $p < 0.05$; −60 min, $F(3,26) = 5.664$, $p < 0.05$; 0 min, $F(3,26) = 4.182$, $p < 0.05$; 30 min, $F(3,26) = 8.544$, $p < 0.05$; 60 min, $F(3,26) = 7.979$, $p < 0.01$; 90 min, $F(3,26) = 4.551$, $p < 0.05$]. In particular, the peak concentration of serotonin during

KCl stimulation in YKS-treated aged rats reached about 2-fold that in young rats.

The basal dopamine concentration (Fig. 4B) was also significantly decreased in aged rats compared to young rats at several time points [−90 min, $F(3,26) = 7.512$, $p < 0.05$; −60 min, $F(3,26) = 7.745$, $p < 0.05$; −30 min, $F(3,26) = 9.598$, $p < 0.05$]. The average dopamine concentrations (in pg/30 μ l, mean $\pm$ SEM) in young and aged rats were as follows: 1.55 ± 0.07 and 0.51 ± 0.05 , respectively. A marked increase in dopamine concentration was observed in young rats under high KCl perfusion; however, this high KCl-evoked increase was significantly attenuated in aged rats compared to young rats [30 min, $F(3,26) = 5.401$, $p < 0.01$; 60 min, $F(3,26) = 11.807$, $p < 0.001$; 90 min, $F(3,26) = 10.741$, $p < 0.001$]. Thus, the dopamine concentration in addition to the serotonin concentration in the PFC was decreased in aged rats. In the drug treatment study, the decreased basal concentration in aged PFC was significantly improved by YKS treatment [−90 min, $F(3,26) = 7.512$, $p < 0.01$; −60 min, $F(3,26) = 7.745$, $p < 0.001$; −30 min, $F(3,26) = 9.598$, $p < 0.001$; 0 min, $F(3,26) = 4.769$, $p < 0.01$]. The attenuated KCl-evoked response was partially but significantly improved by the treatment [30 min, $F(3,26) = 5.401$, $p < 0.05$; 60 min, $F(3,26) = 11.807$, $p < 0.01$; 90 min, $F(3,26) = 10.741$, $p < 0.01$].

3.4. Plasma concentrations of AST, ALT, and NH_3

Plasma concentrations of AST, ALT, and NH_3 were measured, and compared among young, aged, and YKS-treated aged rats (Fig. 5). AST concentrations did not differ among three experimental groups. ALT concentrations were significantly increased in aged rats compared to young rats [$F(2,35) = 9.257$, $p < 0.05$], and this was further increased by YKS treatment [$F(2,35) = 9.257$, $p < 0.01$], but the difference between aged and YKS-treated aged rats was not significant. NH_3 concentrations were also significantly increased in aged rats compared to young rats [$F(2,35) = 10.025$, $p < 0.001$], and this was significantly improved by the treatment [$F(2,35) = 10.025$, $p < 0.001$].

4. Discussion

In the present study, aged rats showed unchanged locomotor activity, but increased excrements during activity measurement (Fig. 1). In addition, aged rats showed less time and decreased frequency in the open arms of the elevated plus-maze (Fig. 2). These results indicate that aging increases anxiety, which is supported by a previous report demonstrating that senescence-accelerated prone mouse 8, a brain aging model, shows increased anxiety with age (Chen et al., 2007).

YKS is effective for a number of symptoms associated with borderline personality disorder, including low mood, impulsivity, and aggression (Miyaoka et al., 2008a), neuroleptic-induced dyskinesia in patients with schizophrenia (Miyaoka et al., 2008b). In addition, this drug is sometimes used to treat BPSD, which includes anxiety, in patients with dementia (Iwasaki et al., 2005; Mizukami et al., 2009; Monji et al., 2009). In animals, YKS has been shown to ameliorate emotional disturbances, such as aggression, in Tg2576 mice expressing the human form of APP695SWE (APP-transgenic mice) (Tabuchi et al., 2009), and Takeda et al. (2008) have demonstrated that this drug suppresses an excessive increase in glutamate release in the hippocampus of zinc-deficient rats, a neurological disease model. These findings led the possibility that this *Kampo* drug can modulate emotion in aged subjects; therefore, we examined the improving effects of YKS on age-related increased anxiety. As shown in Figs. 1 and 2, YKS improved the anxiety-related parameters observed in aged rats, suggesting that YKS has anxiolytic activity against anxiety enhanced by aging.

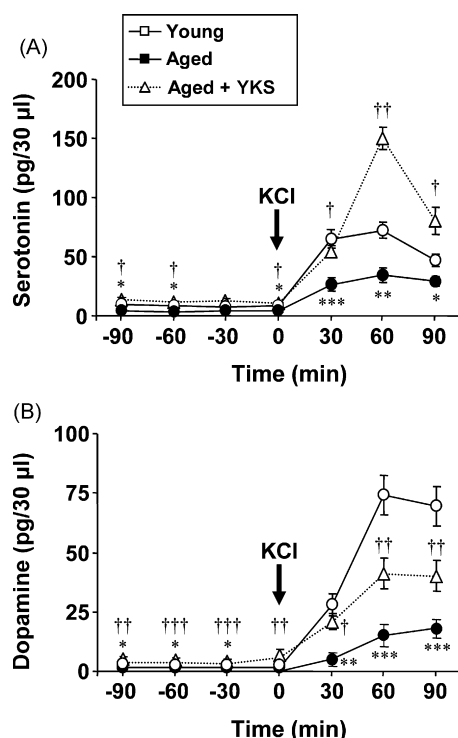


Fig. 4. Extracellular concentrations of serotonin (A) and dopamine (B) in the PFC under basal and KCl-stimulated conditions in young, aged, and YKS-treated aged rats. Each point is the mean $\pm$ SEM of 8–9 rats per group. Asterisks indicate a significant difference from young control rats, * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$. Daggers indicate a significant difference from aged control rats, † $p < 0.05$, †† $p < 0.01$, and ††† $p < 0.001$.

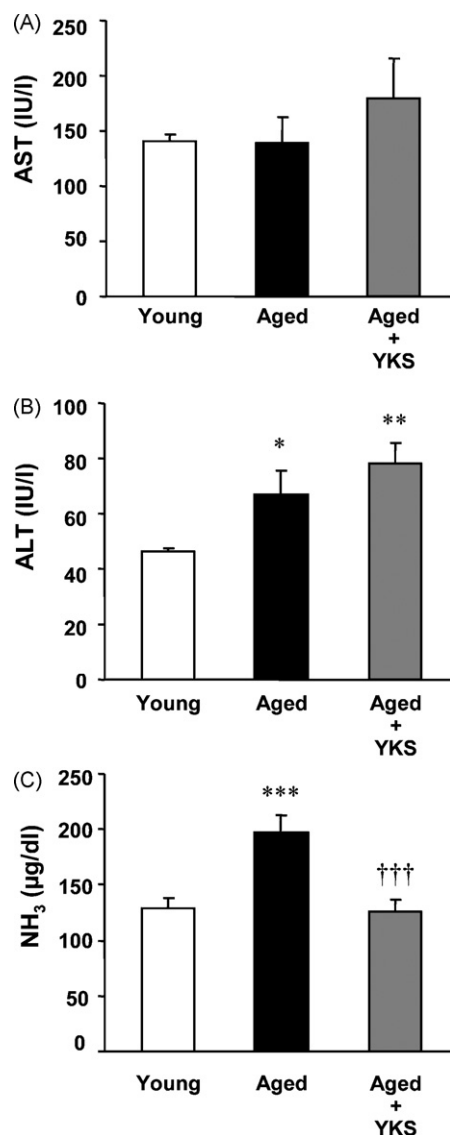


Fig. 5. Influence of aging and effect of YKS treatment on the plasma concentrations of aspartate aminotransferase (AST, A), alanine aminotransferase (ALT, B), and ammonia (NH₃, C). Each column is the mean $\pm$ SEM of 10–18 rats per group. Asterisks indicate a significant difference from young control rats, * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$. Daggers indicate a significant difference from aged control rats, ††† $p < 0.001$.

Although many types of neurotransmitters are associated with regulating anxiety, serotonin and dopamine have an important role; for example, several reports have shown that genetic modulation of the serotonergic system affects anxiety (Gross et al., 2002; Toth, 2003), and a regulatory mechanism of the serotonergic system in several brain regions, including the basolateral and central amygdaloid nuclei, bed nucleus of the stria terminalis, dorsomedial hypothalamus, and PFC, is thought to be involved in anxiety (Gray and McNaughton, 2000; Millan, 2003; Walker et al., 2003; Evans et al., 2006). In addition, reduced dopaminergic activity in the striatum and PFC with 6-hydroxydopamine causes anxiogenic response in the elevated plus-maze test (Tadaiesky et al., 2008), and Watt et al. (2009) have suggested that decreased dopamine levels in the medial PFC is involved in increased anxiety induced by social defeat stress. Moreover, reduced PFC dopaminergic function or the blockade of dopamine receptors in the PFC impairs working memory (Brozoski et al., 1979; Simon et al., 1980; Bubser and Schmidt, 1990). With advancing age, there is marked loss of

dopamine and its metabolism in the PFC of rats (Mizoguchi et al., 2009b), monkeys (Goldman-Rakic and Brown, 1981; Wenk et al., 1989), and humans (Ota et al., 2006), and both PFC cognitive deficits and PFC dopamine depletion emerge early in the aging process, and progressively worsen as aging progresses (Wenk et al., 1989; Bachevalier et al., 1991). Given the importance of serotonin and dopamine in the PFC to regulate emotion and cognition, we next examined the influence of aging and the improving effects of YKS on serotonergic and dopaminergic transmissions in the PFC. As shown in Fig. 4, aged rats showed decreases in extracellular concentrations of serotonin and dopamine in the PFC, suggesting that aging reduces serotonergic and dopaminergic functions in the PFC. The data on decreased dopamine concentration are consistent with our recent report (Mizoguchi et al., 2009b). In the drug treatment study, YKS improved the age-related dysregulation of both neuronal activities, but the effects on dopamine concentration were partial. Thus, the improving effects of YKS on the age-related serotonergic and dopaminergic dysregulations in the PFC were thought to be involved in its anxiolytic activity in aged rats.

However, the mechanisms underlying the improving effects of YKS on serotonergic and dopaminergic dysregulation in the aged PFC, and the active compound(s) causing these effects are unknown. Previous reports have shown that either an increase in serotonin release by inhibition of its transporter at the presynaptic site or stimulation of serotonin receptors with their agonist in the PFC increases dopamine release there (Tanda et al., 1994; Matsumoto et al., 1999). Therefore, the improvement of dopaminergic dysregulation might be due to increased serotonergic transmission by YKS, but other mechanisms are unclear.

Interestingly, YKS improved age-related increase in plasma concentrations of NH₃, but not AST and ALT (Fig. 5), suggesting that YKS enhances age-related reduction in detoxification ability in the liver, but might not improve hepatocyte degeneration. Thus, YKS has improving actions for both anxiety and a part of liver dysfunction in aged rats. This finding is in line with the traditional theory that YKS improves the *Kampo* concept “liver disease” that means emotional disturbances and dysfunction of the liver itself. Considering that NH₃ shows neurotoxicity and that sustained elevation of plasma NH₃ levels can increase glutamate levels in the brain, which in turn causes brain dysfunctions involving a spectrum of neuropsychiatric and neurological symptoms (impaired memory, shortened attention span, sleep–wake inversions, brain edema, intracranial hypertension, seizures, ataxia and coma) (Bosoi and Rose, 2009), the “liver disease” concept is thought to also be reasonable in Western medicine, and it is possible that YKS exhibits its pharmacological potency through the brain–liver interaction.

Anxiety is the most frequent psychiatric condition, and more than 20% of the adult population suffers anxiety at some time during their life. In addition, from 2001 to 2026, the global population aged 65 years and older is predicted to increase from 550 to 973 million in the world (Kinsella and Velkoff, 2001). The elderly population is at risk for various clinical disorders, and anxiety is increasing in the elderly. Benzodiazepine derivatives and SSRIs are used to treat acute anxiety states; however, because of the limited efficacy of the current drugs, the medical need for newer, better-tolerated and more efficacious treatments remains high. Herbal therapies can be considered alternative/complementary medicines, and the search for novel pharmacotherapy from medicinal plants for psychiatric illnesses has progressed significantly in the past decade (Zhang, 2004). Traditional Japanese (*Kampo*) and Chinese medicines have tried to improve a large number of neuropsychiatric disorders in the elderly as well as the young and adults for many years. The present findings help to accumulate scientific evidence of the pharmacological actions of YKS for age-related anxiety.

5. Conclusion

The present data indicate that YKS shows improving action for increased anxiety and for serotonergic and dopaminergic dysregulations in the PFC in aged rats. This mechanism may underlie the effectiveness of this *Kampo* drug for emotional disturbances frequently observed in neurodegenerative disorders, including Alzheimer's disease. In addition, the data suggest that YKS enhances age-related reduction in detoxification ability in the liver. Thus, this paper provides partial scientific evidence for the traditional concept of *Kampo* medicine, "liver disease".

Conflict of interest

None of the authors have any actual or potential conflicts of interest.

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